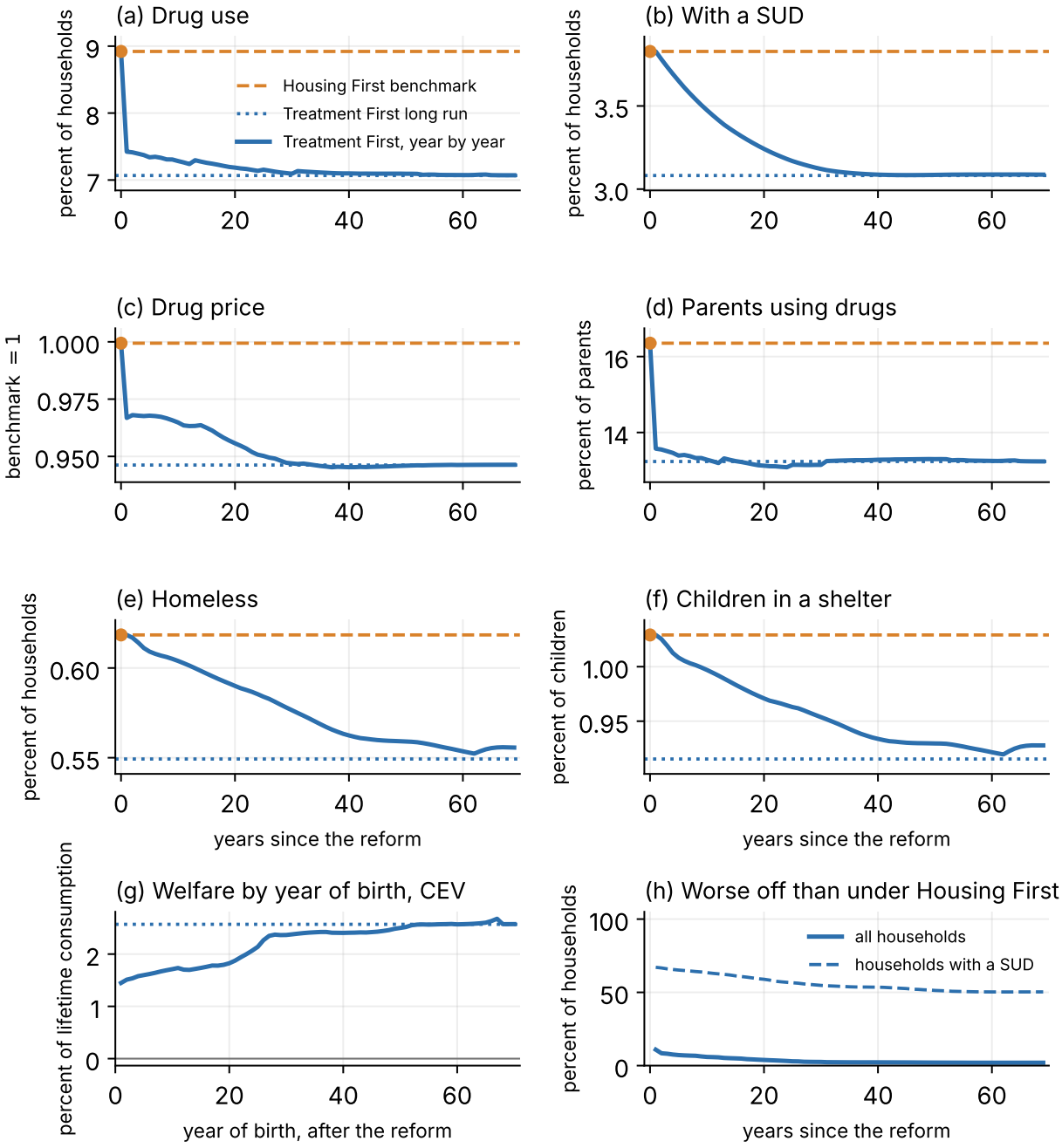


Figure 2: The transition from Housing First to Treatment First, quantitative model

Notes: panels (a) to (f) and (h) are by year since the reform, which lands at the start of year 1, with the benchmark as year 0; the dashed orange line is the Housing First benchmark, the dotted blue line the Treatment First long run and the solid blue line the path. In panel (h) the dashed line is the share of households with a SUD who are worse off. Children are those of households aged 26 to 50. Panel (g) is the CEV of each cohort born on the path, against the benchmark, by its year of birth. Panel (h) is the share of households worse off in each year of the path than in the same state under Housing First, weighted by the benchmark's distribution.

so that only after recovery can they stay sober and pass the screen. I consider three alternatives, each with the screen catching every user or catching 20 percent of users, and with the drug price and π_V clearing every period. In the first, the screen applies from day one of the reform, and the disorder fails it whether or not the household is in treatment. In the second, the reform is announced, but Housing First continues for five periods and the same screen starts in period six. In the third, the same screen applies from the reform, but for its first five periods a household in treatment keeps its voucher even if it uses drugs. Table 7 shows how homelessness, the share of children in a shelter and the welfare of children born at the reform compare with the Housing First benchmark in each, beside the main transition (first row).

Table 7: The first years of the alternative transitions

		Homelessness		Children in a shelter		Newborn's
	p_{det}	Largest rise	Periods above	Largest rise	Periods above	gain
Main transition	20	none	none	none	none	1.45
No exemption						
	100	13.6 (2)	1–15	10.8 (2)	2–11	6.74
	20	–	–	–	–	–
Announced five periods ahead						
	100	7.3 (9)	7–16	3.6 (9)	7–12	4.94
	20	3.1 (12)	7–19	0.6 (11)	9–13	2.57
Sobriety while in treatment exempt for the first five periods						
	100	4.2 (9)	7–13	0.2 (9)	8–9	6.88
	20	1.5 (11)	9–15	none	none	3.14

Notes: Each row is a transition from the Housing First benchmark to Treatment First with the drug price and π_V clearing every period, against Housing First's steady state. Main transition: Section 6.4, where a household with a SUD keeps its voucher by entering treatment and not using. In the other rows it cannot stop using, and the disorder fails the screen even in treatment: from the reform (no exemption), from period six with Housing First until then (announced five periods ahead), or from period six with treatment alone passing it until then (sobriety while in treatment exempt for the first five periods). p_{det} is the detection rate, in percent. Largest rise: the largest percent increase over the first fifteen periods, with its period in brackets. Periods above: the periods in which it is above Housing First. Children in a shelter is the homelessness of households aged 26 to 50. Newborn's gain: the CEV of children born at the reform, percent of lifetime consumption. When every user is caught, no drug price clears the long-run drug market, because drug use jumps as the price moves through the level that would clear it; those transitions end at the long-run solution that comes closest.

In the first alternative, how much the screen catches decides what happens along the way. Catching every user lowers homelessness by 30 percent and the share of children in a shelter by 31 percent in the long run, but it puts every voucher holder with a SUD out at once: homelessness rises by up to 14 percent and the share of children in a shelter by up to 11 percent, and they stay above their Housing First levels for 15 and 11 years, respectively, because π_V rises only as the savings accrue. Catching only one in five users lowers both in the long run too, homelessness by 11 percent and the share of children in a shelter by 11 percent, but it also raises them in the first years, though by less, because it puts fewer households out at once: before its savings are recycled and the drug price falls, homelessness rises by up to 6 percent and the share of children in a shelter by up to 4 percent. Even so, children born at the reform gain 6.74 percent of lifetime consumption when every user is caught and – percent when 20 percent are, against 1.45 percent in the main transition. They gain more because the screen deters more: it catches every user, and a household with a SUD cannot stop using, so the disorder costs it its voucher until it recovers, while in the main transition the household keeps the voucher by entering treatment and not using. By period ten, 2.80 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder, and π_V is higher throughout their childhood.

Announcing the screen five periods before it starts lowers the rise but does not eliminate it. While Housing First continues, households cut their drug use at once in anticipation of a screen that will catch every user, by 31 percent in the first period, and homelessness and the share of children in a shelter fall below their Housing First levels. Once it starts revoking vouchers, they rise by up to 7 and 4 percent,

against 14 and 11 percent without the announcement. When it catches one in five users, they rise by up to 3 and 0.6 percent. Children born at the reform gain 4.94 percent of lifetime consumption when every user is caught and 2.57 percent when one in five users are. The first is less than the 6.74 percent without the announcement, because π_V does not rise until the screen starts and drug use falls by less in the first period than the 40 percent when the screen starts at once. The second is also more than the 1.45 percent in the main transition, although the screen catches one in five users in both. Here a household with a SUD cannot stop using, so once the screen starts the disorder puts its voucher at risk until it recovers, while in the main transition the household keeps the voucher by entering treatment and not using. Starting to use therefore risks more, and fewer households use drugs: by period ten, 3.27 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder.

Exempting sobriety while in treatment for the first five periods lets a household with a SUD keep its voucher in those periods by entering treatment, even if it uses drugs, and more of them do: 36 percent of those with a SUD are in treatment, against 25 percent under Housing First. When the screen catches one in five users, homelessness stays below its Housing First level for the first eight periods but rises above it by at most 1.5 percent after the exemption ends, and the share of children in a shelter never rises above its Housing First level. When every user is caught, homelessness stays below its Housing First level for the first six periods but rises above it by at most 4.2 percent after the exemption ends, and the share of children in a shelter by at most 0.2 percent. Children born at the reform gain 3.14 percent of lifetime consumption when one in five users are caught, more than the 1.45 percent in the main transition, because once the exemption ends a household with a SUD, which cannot stop using, risks its voucher until it recovers, so fewer households use drugs: by period ten, 3.22 percent of households have a SUD, against 3.47 percent, so fewer of these children grow up with a parent who has the disorder, and π_V is higher throughout their childhood. When every user is caught, they gain 6.88 percent.

7 Discussion and implications

To close, I return to the question I started with: should housing assistance require sobriety? The answer turns on the children, and not on the moral hazard that dominates the policy debate: without the externalities on children, Treatment First cuts drug use by 28 percent, but a newborn is worse off, because the screen takes an option from households that weigh the risks correctly and gives back less through the budget than it takes.

Drug use costs children consumption and housing at home and productivity in adult life, and a parent counts only the share ω_c of either, so the screen's deterrence is worth more to the child than to the parent. With the drug price and π_V held, the parents lose 1.19 percent of their own lifetime consumption, about what they lose without the externalities, while a newborn gains 2.82 percent, so the reform raises welfare on net once the children are counted. With the budget rebalanced and the drug price clearing, as they would be after a real reform, Treatment First cuts drug use by 21 percent, lowers homelessness by 11 percent, and raises a newborn's welfare by 2.57 percent of lifetime consumption, three quarters of it an externality its parent does not count. Once general-equilibrium effects are accounted for, the parents gain too, because the higher π_V and the cheaper drug outweigh the option the screen removes. Along the transition, neither homelessness nor the share of children in a shelter rises, and children born at the reform gain 1.45 percent.

Three lessons follow. First, the arguments for Housing First and Treatment First made by different advocacy groups leave out the outcome that matters most: what happens to the children of the households a sobriety requirement deters from using, in what they consume while living with their parents, in whether they become homeless and in what they earn as adults.

Second, the model agrees with the micro evidence behind Housing First on the households it follows: housing, and a sobriety condition on it, raise the recovery of households with a SUD by about as much as in the trial that measures both, and losing a voucher raises the mortality of households with SMI by about as much as Housing First lowers it among homeless veterans with mental illness. As that evidence implies, those with a SUD lose from Treatment First: with the drug price and π_V held, as in a trial, which is too small to move either, – percent of those who have one when the reform lands are worse off, and even once both clear, 67 percent are. The macro effects the trials do not capture, however, overturn the conclusions.

The trials follow only households already in a bind: adults who are homeless, most with SMI and many with a SUD. They do not follow the households that could get into a bind but that the condition deters from starting to use drugs. Nor do they observe the vouchers the condition's savings buy for others, or the children's adult lives. Once those are counted, aggregate homelessness and childhood homelessness drop and welfare increases.

Third, homelessness and childhood homelessness do not fall along the transition if treatment and sobriety are not a real option for the households that already have a SUD. A temporary exemption for those with a SUD keeps the latter from ever rising and the former's rise small: if a household in treatment keeps its voucher, sober or not, for the first five periods, and the screen catches one in five users, the share of children in a shelter never rises above its Housing First level and homelessness rises by at most 1.5 percent, with newborns gaining even more than in the main transition.

A Calibration of the standard parameters

The remaining thirty-three parameters are more conventional, and I record the values I use so that the calibration can be reproduced. Six of them I calibrate internally, against a moment, because no source gives them directly; the other twenty-seven I set from outside, from an estimate or from statute. Table A1 and the entries below run in the same seven blocks, and each entry gives the moment or the source that fixes the parameter together with the value that moment takes in the data. The Model column reports what the model produces against that value, and it does so only for the six I calibrate internally, since the other twenty-seven have no target to reproduce.

The parameters in this appendix are not inert in the experiments. For example, the discount factor governs how heavily a household weighs a death later in life, the dispersion of productivity determines how many households are poor enough to be at risk of homelessness and of acquiring a disorder, and risk aversion governs what insurance is worth. What separates them from the parameters of Section 5 is not that they do nothing, but that they are not specific to this paper.

Table A1: Standard parameters

Parameter	Description	Set	Moment and source	Data	Model	Value
Block 7: Preferences						
σ	Risk aversion	Ext.	Attanasio and Weber (2010)	2.00	n.a.	2.00
γ	Weight on housing services	Int.	Housing over expenditure net of drugs, CEX 2024	26.8%	26.2%	0.536
β	Discount factor	Int.	Wealth over output, BEA fixed assets	3.00	2.95	0.946
$\xi(j)$	Equivalence scale	Ext.	Square-root scale, OECD	1.41	n.a.	1.41
Block 8: Rental ladder						
h_2	Smallest rental size	Int.	Renters paying over half of income, SCF 2022	19.2%	25.8%	0.0407
h_3/h_2	Middle rung over the smallest	Ext.	Set	4.00	n.a.	4.00
h_4/h_2	Largest rung over the smallest	Ext.	Set	8.00	n.a.	8.00
Block 9: Earnings and employment						
J	Maximum age	Ext.	Age 85	60	n.a.	60
J_K	Years the child lives at home	Ext.	Parent aged 26 to 50	25	n.a.	25
J_R	Retirement age	Ext.	Age 66, SSA full retirement age	41	n.a.	41
$\varepsilon(j)$	Age profile of efficiency	Ext.	Hump peaking at age 56, Greenwood et al. (2025)	Profile	n.a.	Profile
ρ_y	Persistence of productivity	Ext.	Guvenen et al. (2023)	0.90	n.a.	0.90
σ_ζ	Standard deviation of its shocks	Int.	Bottom half's share of earnings, SCF 2022	17.2%	16.2%	0.413
ρ	Persistence of productivity across generations	Ext.	Solon (1992), Mazumder (2005)	0.50	n.a.	0.50
$\bar{\pi}^{eu}$	Job separation rate	Int.	Unemployment rate, BLS CPS 2014–24	4.9%	5.6%	0.0268
$\varpi(y)$	Separations by productivity	Ext.	Unemployment by education, BLS 2019	Profile	n.a.	Profile
$\bar{\pi}^{ue}$	Annual job-finding rate	Ext.	Krueger et al. (2014)	0.55	n.a.	0.55
p_{ui}	Share of separations that receive UI	Ext.	UI applicants among the unemployed, BLS 2022	0.30	n.a.	0.30
Block 10: Baseline survival and SMI onset						
$\bar{\phi}(j)$	Baseline survival by age	Ext.	US life table 2010, Arias (2014)	Profile	n.a.	Profile
$\varsigma_m(j)$	Age profile of SMI onset	Ext.	SMI prevalence by age, NSDUH 2024	Profile	n.a.	Profile
Block 11: Interest and taxes						
r	Risk-free rate	Ext.	Kaplan et al. (2020)	0.03	n.a.	0.03
φ	Tax progressivity	Ext.	Heathcote et al. (2017)	0.85	n.a.	0.85
τ_ℓ	Level of the labour income tax	Int.	Average labour income tax, McDaniel (2007)	22.4%	20.9%	0.894
τ_k	Capital income tax rate	Ext.	US long-run rate on gains and dividends	0.15	n.a.	0.15
Block 12: Social security benefits and programmes' rules						
$S(y)$	Social security	Ext.	SSA primary insurance amount	Statute	n.a.	0.90, 0.32, 0.15
$\underline{S}$	SSI floor over average labour income	Ext.	SSA federal benefit rate	0.177	n.a.	0.177
ψ_{ui}	UI replacement rate	Ext.	Department of Labor, statutory rates	0.50	n.a.	0.50
$\bar{y}$	Cap on the UI earnings base	Ext.	Department of Labor, state benefit maxima	1.00	n.a.	1.00
κ_V	Voucher means test	Ext.	HUD rule	0.50	n.a.	0.50
κ_r	Tenant's rent, share of income	Ext.	HUD rule	0.30	n.a.	0.30
ϱ	Food stamp clawback rate	Ext.	SNAP statute, 7 CFR 273.10	0.30	n.a.	0.30
Block 13: Drug market						
J_E	Maximum age at which use can begin	Ext.	Model age 55, that is age 80	55	n.a.	55
ν_1	Supply shifter	Ext.	Normalisation, so $p_v = 1$	Norm.	n.a.	Norm.

Notes: the Model column is the Housing First benchmark. n.a. marks a parameter set from outside the model, which has no moment of its own to reproduce.

Block 7: Preferences. The four parameters in this block determine what a household wants, and what it is willing to give up to get it.

σ, risk aversion. It sets the marginal utility of consumption, so it affects every welfare comparison in the paper. Increase it and households want smoother consumption and insurance is worth more, the voucher included. I set it to the standard value of two, following Attanasio and Weber (2010).

γ**, the weight on housing services.** It is the weight on housing in the bundle of consumption and housing, so it sets what a dwelling is worth against consumption, and that bundle is what the drug is weighed against. As γ rises housing is worth more relative to consumption, households buy more of it, and the share of spending that goes on housing rises with it. γ is chosen for the share it produces: shelter plus utilities take 26.8 percent of expenditure in the 2024 Consumer Expenditure Survey (CEX), which excludes illicit drug spending.

β**, the discount factor.** It sets how much households discount the future, and thus how much they save and how much the prospect of death deters them from using drugs. As it rises households save more, the wealth-to-output ratio rises and households are more deterred by death. It is chosen so that wealth over output matches the national accounts, a ratio of three, from the Bureau of Economic Analysis (BEA) fixed asset tables against National Income and Product Accounts (NIPA) output, both read at the 2023 vintage, the last year the fixed asset tables cover.

$\xi(j)$**, the equivalence scale.** The household's consumption and dwelling are shared with the child while it lives with its parent, and thus are divided by $\xi(j)$ to give what each member gets. Raise it and each member gets less of a given budget, and the child counts for more of the household, through the weight $\xi(j) - 1$ in (12). It is the square-root scale (OECD, 2013), $\sqrt{N(j)}$, where $N(j)$ in the model is set to two while the child lives with the parent ($j \leq J_K$), from the parent's age 26 to 50, and to one after the child has left ($j > J_K$).

Block 8: Rental ladder. These three parameters build the grid of dwelling sizes a household chooses from. Below the smallest rung is the shelter, h_1, which was calibrated in Block 2.

h_2**, the smallest rental size.** The size of the cheapest dwelling is what decides how much of its income a poor household must spend to be housed at all. Raise it and the bottom rung is larger and dearer, so more households are burdened, and a voucher is worth more to them. It is chosen so that the fraction of renters paying more than half of their income for housing is similar to the data, 19.2 percent in the 2022 SCF; the model gives 25.8 percent.

h_3/h_2 **and** h_4/h_2**, the spacing of the ladder.** These two ratios set how much housing the middle and largest rungs provide, and so what a household that can afford more than the smallest unit can move up to. Raise them and each step up the ladder costs more rent, and fewer households can afford the larger rungs. The two larger rungs are set at 4.00 and 8.00 times the smallest.

Block 9: Earnings and employment. These eleven parameters determine how long a household lives and works, what it is expected to earn over its life, how persistent the shocks to it are, and how it moves between work and unemployment.

J**,** J_K **and** J_R**, the length of life, of child rearing and of working life, respectively.** A household lives at most J periods, has its child at home for its first J_K of them and works until model age J_R, after which it retires on social security. Together they set how many years of life and of earnings a disorder can cost, and J_K how long a parent's drug use reaches its child through what they share. Raise either J or J_R and a disorder that shortens life or work costs more; raise J_K and the *budget and childhood homelessness externalities* last longer. I set $J = 60$, so that with $j = 1$ at age 26 the last age is 85, $J_K = 25$, so that the child lives at home while its parent is 26 to 50, and $J_R = 41$, so that retirement begins at age 66, the full retirement age under social security for those born between 1943 and 1954.

$\varepsilon(j)$**, the age profile of efficiency.** This is the deterministic hump in earnings over the life cycle. Without it every household with the same productivity would earn the same at every age, and the ages at which households are poor, and so at risk of homelessness and of using, would be wrong at both ends of working life. I take it as the exponential of a quadratic in age used by Greenwood et al. (2025), $\varepsilon(j) = \exp\{-(j-1)^2/1800 + (j-1)/30\}$ with $j = 1$ at age 26. It peaks at age 56, having risen 65 percent from age 26, and then falls 4 percent from the peak to age 65, the last working age.

ρ_y **and** σ_ζ**, the productivity process.** Log productivity is persistent but not permanent: it follows an autoregressive process with persistence ρ_y and shocks with standard deviation σ_ζ. σ_ζ is what decides how many households are poor enough to be at risk of homelessness, and the share of earnings that

goes to the bottom half of working households falls as it rises. I set $\rho_y = 0.90$, the persistence of the income process in Guvenen et al. (2023). σ_ζ is calibrated to the bottom half's share of earnings among households aged 26 to 65 with positive wage income in the 2022 SCF, 17.2 percent.

ρ**, the persistence of productivity across generations.** The child's log productivity at entry into adulthood is drawn around its parent's with weight ρ and around the mean with weight $1 - \rho$, so ρ is how much of its parent's current productivity a child keeps. It is the channel through which anything that lowers a parent's productivity lowers its child's: a parent whose own productivity carries the penalty δ_d, because its own parent had a SUD, also passes a lower draw on to its child, so a parent's SUD lowers productivity beyond the next generation. Raise it and a child's prospects depend more on its parent's decisions, and the *earnings externality* grows. It is set externally at 0.50, between the intergenerational elasticities of earnings in Solon (1992) and Mazumder (2005).

$\bar{\pi}^{eu}$**, the job separation rate.** This parameter determines the chance that an employed household loses its job each year. It is scaled by $\varpi(y)$ in (21), so separations are likelier at low productivity. Raise it and more households lose work each year, so the unemployment rate rises with it. It is chosen to reproduce the aggregate unemployment rate in the Current Population Survey (CPS) of the Bureau of Labor Statistics (BLS) over 2014 to 2024, 4.9 percent, given the job-finding rate set externally below.

$\varpi(y)$**, separations by productivity.** This parameter tilts the separation rate across productivity, so that low-productivity households lose their jobs more often, which is what puts them at a higher risk of homelessness. Steepen the tilt and job loss concentrates further among the poorest, and so does the risk of homelessness. It is normalised to average one across households, so that $\bar{\pi}^{eu}$ alone sets the aggregate unemployment rate. Each productivity node takes the unemployment odds of the education group it falls in, from the BLS 2019 annual averages of unemployment by educational attainment for ages 25 and over, with the education groups ordered along productivity by their median earnings.

$\bar{\pi}^{ue}$**, the annual job-finding rate.** This parameter is the probability that an unemployed household finds work each year it is unemployed. With the divisors κ_d^{ue} and κ_m^{ue} it sets how long an unemployed household with a SUD or SMI is expected to remain unemployed. Raise it and spells out of work are shorter on average, so fewer households fall behind on the rent. I set it to 0.55 from the annual exit rate Krueger et al. (2014) estimate on matched CPS panels.

p_{ui}**, the share of separations that receive UI.** A household that loses its job draws UI for one year with probability p_{ui} and nothing otherwise. It sets how many of the newly unemployed fall straight to zero earnings, and how costly being unemployed is. I set it to 0.30, since in the BLS supplement to the CPS for 2022 about seven in ten unemployed people who worked in the past year did not apply for UI after losing their job.

Block 10: Baseline survival and SMI onset. These two parameters are the survival and the SMI-onset age profiles.

$\bar{\phi}(j)$**, baseline survival by age.** This is the baseline chance of surviving to the next age, before any multiplier. It is the base that every mortality multiplier in Section 5, for a SUD, homelessness and SMI, scales. Raise it and households expect to live longer, so the life a SUD puts at risk is worth more. I take the profile from the life table for the total population of the United States in 2010 (Arias, 2014), for ages 26 to 85.

$\varsigma_m(j)$**, the age profile of SMI onset.** This determines the shape of onset across ages, normalised to one at age 26, so that ω_m in Table 3 sets its level alone. It is here rather than in Table 3 because, like the age survival profile, it only sets the baseline. What actually affects the impact of Housing First versus Treatment First is ι, the multiplier a SUD puts on SMI onset, which is in Table 3. It is the onset that reproduces a log-linear fit to the share of adults carrying SMI at each age in NSDUH 2024.

Block 11: Interest and taxes.

r**, the risk-free rate.** This parameter sets the return on saving. Raise it and saving pays more, so households hold more assets to carry them through a spell out of work or without a voucher. I set it to three percent, from Kaplan et al. (2020).

φ**, tax progressivity.** This is the exponent on income in the tax schedule of Section 4.3, and it determines the progressivity of the labour income tax. Lower it and the tax is more progressive, so households with low incomes, among them most voucher holders and most drug users, keep more of what they earn. It is set to 0.85, in line with Heathcote et al. (2017).

τ_ℓ**, the scale of after-tax labour income.** Given φ, this parameter sets how much of their labour income households keep after tax on average, so raising it lowers the tax liability. It does not move in the experiments: the budget is balanced by π_V instead. It is calibrated to an average labour income tax rate of 22.4 percent, which is what McDaniel (2007) measures for the United States and what Guvenen et al. (2023) use.

τ_k**, the capital income tax rate.** This parameter affects the after-tax return on savings. Raise it and saving pays less, so households hold fewer assets to fall back on. I set it to fifteen percent, consistent with the statutory United States rate on long-term capital gains and qualified dividends for the middle of the income distribution, which the Internal Revenue Service (IRS) sets out in Topic 409 (`https://www.irs.gov/taxtopics/tc409`).

Block 12: Social security benefits and programmes' rules.

$S(y)$ **and** $\underline{S}$**, social security and its floor.** Retirement income follows the statutory primary insurance amount: it replaces a share of the household's income at its last working age, a share that falls at each of the two bend points of the Social Security Administration (SSA), up to the taxable maximum, with the SSI payment as a floor (the full function is detailed in footnote 16). As $S(y)$ rises the old are richer, so the value of a life still to be lived rises and with it the cost of a disorder that shortens it. The shares $\alpha_1, \alpha_2, \alpha_3$ are $0.90, 0.32, 0.15$ below the first bend point, between the two and above the second, respectively. The bend points k_1 and k_2 and the taxable maximum $y^{\max}$ are the SSA's 2024 dollar amounts divided by its average wage index, 0.22, 1.33 and 2.64, respectively. The floor, $\underline{S}$, is 0.18 of average labour income.

ψ_{ui} **and** $\bar{y}$**, UI.** These two set the UI replacement rate and the cap on the earnings it is paid on, respectively. Raise either and the first year out of work costs a household less, so a spell out of work is less likely to end in a shelter. I set them to fifty percent and one times average labour income, respectively, taking both from the statutory replacement rates and state benefit maxima that the Department of Labor compiles in its comparison of state UI laws (`https://oui.doleta.gov/unemploy/statelaws.asp`).

κ_V **and** κ_r**, the voucher rules.** These two set the income cutoff for eligibility and the tenant's rent contribution, respectively. Raise κ_V and more households are eligible, so more have a voucher to lose under Treatment First; raise κ_r and a voucher pays less, so losing one costs less. I take both from HUD's own rules: half of median income and thirty percent, respectively. The first is the very low-income limit HUD publishes annually with its income limits (`https://www.huduser.gov/portal/datasets/il.html`); the second is the share of its before-tax income a tenant pays towards the rent, the voucher paying the rest up to the payment standard, under the Housing Choice Voucher programme (`https://www.hud.gov/helping-americans/housing-choice-vouchers`).

ϱ**, the food stamp clawback rate.** This is the rate at which food assistance is withdrawn as income rises. Raise it and food assistance is clawed back faster, so it stops at a lower income and pays less to every household that still receives it. I set it to thirty percent, which is the statutory rule (7 CFR 273.10).

Block 13: Drug market. The two parameters in this block describe the rest of the drug market of Block 6. The first, J_E, fixes who can enter the market (footnote 21); the second is set so that supply equals demand in the benchmark, and is then held fixed.

J_E**, the last age at which use can begin.** This parameter is the last age at which a household without a SUD can start using; one that already has a SUD at that age can keep using. It decides who can enter the drug market, and so how many households a policy that deters use can reach. As it rises, more old households can start using at little cost, since too little life remains for the disorder to damage (footnote 21), and the model produces too much old-age use. I set it to model age 55, the equivalent of age 80 when the model runs to age 85.

ν_1**, the supply shifter.** A pure normalisation, chosen so that supply equals demand at $p_v = 1$ in the benchmark, and kept fixed across experiments, in which the price is free to adjust. It normalises units, not behaviour.

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